



Birla Institute of Technology & Science, Pilani

Hyderabad Campus

1st Semester 2025-2026 Course Handout (Part II)

Date: 01/08/2025

In addition to Part-I (General handout for all courses appended to the timetable) this portion gives further specific details regarding the course:

Course No. : CS F407
Course Title : Artificial Intelligence
Instructor (I/C) : **Abhijit Das**

1. Scope

This course introduces students to basic concepts and methods of artificial intelligence from a computer science perspective. AI concerns itself with a certain set of problems and develops a particular body of techniques for approaching these problems. The focus of the course will be on the study of methods of knowledge representation, reasoning, and algorithms required for the developing intelligent systems and programs.

Course Objectives

- Empower students to know how to program computers, using classical symbolic methods, to behave in ways normally attributed to "intelligence" when observed in humans.
- To have an understanding of the core topics in AI such as learning, natural language processing, agents and robotics, expert systems, and planning.
- To have a basic proficiency in a traditional AI language and logic, including the ability to write simple to intermediate programs and understand code.
- Emphasize the use of MATLAB and Python to implement the use of Search strategies in real world problem solving, Game playing programs like chess or tic-tac-toe, Planners, Small Expert system shell with only inference engine, Programs for reasoning under uncertainties etc.
- Cultivate an interest in the field, sufficient to handle more advanced projects.

2. Text Book

T1: Stuart Russell, and Peter Norvig, "Artificial Intelligence: A Modern Approach", Pearson education, 3rd Ed, 2009.

3. Reference Books

R1- George F. Luger "Artificial Intelligence: Structures and Strategies for Complex Problem Solving", Fourth Edition, Pearson, 2002.

R2- D. W. Patterson, "Introduction to Artificial Intelligence & Expert Systems", PHI, 2002.

4. PLAN OF STUDY:

S. No.	Learning Objectives	TOPIC	CHA. REF.	Hrs
1.	To understand need of AI and what can be called as an AI technique.	Fundamental Issues in Intelligent Systems: Definitions, Attitude towards intelligence & knowledge; Agents, Percepts, Environments; Example of an AI Technique.	T1(1), R1(1) Lecture Notes.	2
2.	Learn state space search for problem solving; Different approaches to search a space like heuristics, blind adversarial search etc will be covered.	Problem Solving using Search Strategies: State Space search: Problem Spaces, Graph Theory, and Strategies for State Space Search. Heuristic Search: Generate & Test, Hill Climbing, Best First, Problem Reduction, Properties of Heuristics like Admissibility, Monotonicity, and Informedness. Adversarial Search (Game Playing): Minimax, Alpha-Beta Cutoffs. Local Search & Optimization: Genetic Algorithms & Particle swarm optimization.	T1(3), R1(3) T1(5), R1(4) T1(6), R1(5) T1 (4) Lecture Notes.	2 2 4 4

3.	To develop systems/models that can infer new information & knowledge from existing ones. Also, what would be few right approaches to represent (store) the knowledge to be processed or used in the reasoning. Handle real world data that is vague/uncertain.	Knowledge Representation and Reasoning: Approaches and Issues, Predicate Logic: Syntax, and Semantics of Propositional and First Order Predicate Logic, Conversion to Clause Form, Deduction, Unification, Resolution based Theorem Proving. Reasoning under Uncertainties: Bayes' Theorem, Bayesian Networks, Decision Theory. Fuzzy Logic & Representation of uncertainty Fuzzy Inference Systems	T1(8,9), R1(2) T1(13, 14,16) R3(1, 2, 4) Lecture Notes.	4 4 4
4.	To understand the state of art research in reasoning systems.	Current Research on Knowledge representation and Reasoning	Elsevier publications	1
5.	To build models/programs that can learn from the past. Understand different machine learning algorithms with applications.	Machine Learning: General Concepts in Knowledge Acquisition & Learning; Decision Tree, Ensemble learning Methods. Explanation based learning. Inductive logic programming. Applications of ML: Speech, Language, Vision, Handwritten digit recognition.	T1(18), R1(9), R5 T1(19) T1(20), R1(10) Lecture Notes.	2 2 2
6.	To learn how to write programs that can make a computer interpret images.	Perception: Introduction, Formation, robotic perception and planning to move.	T1(25)	2
7.	To learn architecture / framework for an expert system.	Expert Systems: Rule based Expert System Architecture.	R1(13) Lecture Notes.	2 2
8	To understand the state of art research and applications in Exp. Sys & Decision-Support Sys.	Current Research on Knowledge representation and Reasoning from Exp. Sys. with App;	Elsevier publications	1
Total Lecture hours				40

5. EVALUATION SCHEME:

S. No.	Component & Nature	Duration	Weightage	Date and Time
1	Mid semester Test (Closed Book)	1.5 hrs.	25%	08/10 - 9.30 - 11.00AM
2.	Coding Assignments (Take Home)	--	25% (12% before mid sem)	TBA
3	Surprise class evaluation (Open Book)	-	5% (5 best out of 7 test) 3% before mid sem	TBA
4.	Comprehensive Exam (Closed Book)	3 hrs	45%	05/12 FN

*40% of the Evaluation will be completed by Mid Semester Grading.

Note: All notices related to the course will be posted in the LMS/ google class room.

6. Makeup Policy: Make-up for mid semester exam and comprehensive exam will be granted only on genuine grounds of sickness (**to be supported by a medical certificate and not a prescription**). There is NO makeup for other evaluation components.

7. Chamber Consultation Hour: Will be announced in the class.

8. Academic Honesty and Integrity Policy: Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.

Abhijit Das
Instructor-in-charge